Dr. Raphael Zufferey

Robotics and micro-technology PhD engineer with extensive experience in mobile autonomous systems

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Education

2020-now **Postdoc**, *University of Seville*, Spain.
Member of the Robotics Vision and Control research group led by Prof. Ollero.

Research: Design and development of novel flapping-wing aerial robots to test and validate our biological understanding of the nature counterparts.

2016-2019 **PhD**, *Imperial College London*, UK.
Imperial College London
Member of the Aerial Robotics Lab research group. Research: *Aquatic Escape of Micro Aerial Vehicles*.

Thesis content: Practical and theoretical investigations into miniature aerial-aquatic robots capable of movement in both air and water. The work required extensive experimental work and resulted in two unique prototypes with distinct locomotion strategies. Supervision by Dr. M. Kovac. Department of aeronautics. (*Defended on 27 sept 2019*)

2015 **Master thesis**, *Harvard University*, US.
HARVARD UNIVERSITY
Student at the Microrobotics Laboratory research group. Research: Development of a micro legged robot of 2.5 grams, the Harvard Ambulatory Micro Robot (HAMR), involving key advances in piezoelectric actuators, micro PCB fabrication, wireless implementation. Significant challenges in small scale system integration were overcome. Supervision by Prof. R.J. Wood and Prof. D. Floreano.

2010-2015 **MSc, BA**, *EPFL*, Switzerland.
EPFL ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE
Micro-engineering bachelor and master, emphasis on electrical, mechanical and material science in addition to micro-systems, from semi-conductor physics to embedded programming.
Specialization in robotics and autonomous system.
Minor in Space Design and Technology
Final grade 5.4/6

< 2009 **Maturité**, *Gymnase de Nyon*, Switzerland, Equivalent of a Baccalaureate. Bilingual FR-DE.

Experience

2019-2020 **Research Scientist**, *Imperial College London*.
Post-doctoral researcher as part of the CASCADE aerial-aquatic research program

2013-2018 **Teaching**, *University*.
Imperial College London: 3rd year robotics application class. Experiment setup and guidance.
EPFL: Physics I, Physics II and Electromechanic Conversion teaching assistant.

2016-2019 **Webmaster**, *Imperial College London*.
Webmaster of the Aerial Robotics Lab website, including internal wiki development.

2014 **Intern**, *Logitech*, Switzerland.
Collaboration with Future Labs in the implementation of gesture control in a professional 2D camera.

2014 **Competition**, *EPFL*.
Autonomous robotics competition, development and construction of a multi-terrain robot which localises itself, detects, picks up and brings home plastic bottles in unknown environment with obstacle avoidance and path planning. (supervision by Prof. A. Ijspeert)

2013-2014 **Semester Projects**, *EPFL*.
Project 1: Finite Element analysis of hyperelastic membranes in DEAs.
Project 2: Digital Sun Sensor : Interfacing and programming of a bare-die HDR camera array.

2009-2010 **Language Development**, *Bath Academy, UK & Bellevue College Seattle, US*.
Intensive english studies and public speaking classes.

Leadership

2016-2019 **Supervisor**, *Imperial College London*.
Close supervision of 3 master students and 2 undergraduate in aerial aquatic robotics. Work ranging from chemical propulsion vehicles, water entry studies and launch systems development to sailing flying robots.

2008-2016 **Scout Group Leader**, *Nyon*.
7 summer camps organized for 10-80 young people. Jeunesse et Sport (J+S) leader licence. Water rescue licenses: Pool, Lake and River Rescue.

2012-2015 **Department Delegate**, *EPFL*.
micro-engineering class delegate at EPFL.

2012-2013, 2018 **Uni societies committee**.
Coaching society, *EPFL*
Micro-technology society, *EPFL*
Windsurfing society, *Imperial College London*

Skills

Professional Micro technologies: analog and digital electronics, mechanics and micro-structure, robotic design, embedded software, semi-conductor physics, machine learning, evolutionary algorithms, FEM

Practical Machining, composites fabrication, 3D printing (plastics/metals), Laser micro machining, PCB design and fabrication, unmanned aircraft design, 3D motion capture system

Computing C, C++, Matlab, Solidworks, Draftsight, GitHub, Extensive Adobe Illustrator/ Photoshop/ Premiere/ Lightroom/ AfterEffect, Blender, LaTeX

Sports Rock climbing 3x/week, windsurf, ski, snowboard, surf, badminton & tennis

Music Classical Piano since 1999 and guitar since 2013

Language **French**, native
German, native
English, proficient, *IELTS, score 8/9*
Spanish, fluent

Awards

2020 *Best UK PhD in Robotics*, one of 3 finalists, results to be announced

2019 *Best Paper Award*, AMAM 2019, Lausanne Switzerland (Only award out of 82)

2018 *Best Conference Paper Nominee*, ICRA 2018, Brisbane

2017 *Winner, Best Robot Video*, 'Aquatic Micro Air Vehicles for Water Health Monitoring', AAAI-17 conference, San Francisco

2009 Swiss Physics Olympiads nominee

Publication Record [ORCID]

Journal Publications

2020 **MEDUSA: a Multi-Environment Dual-robot for Underwater Sample Acquisition**.
RA-L, Robotics and Automation Letters Article Video
D. Debruyn, R. Zufferey, *SF. Armanini, C. Winston, A. Farinha, Y. Jin and M. Kovac*

2020 **Unmanned Aerial Sensor Placement for Cluttered Environments**.
RA-L, Robotics and Automation Letters Article
A. Farinha, R. Zufferey, *P. Zheng, SF. Armanini and M. Kovac*

2019 **Multiple Consecutive Jet-gliding Flights with Water-reactive Fuel**.
Science Robotics Article Video
R. Zufferey, *A. Ortega Ancel, A. Farinha, [...] and M. Kovac*

2019 **SailMAV: design and implementation of a novel multi-modal flying sailing robot**.
RA-L, Robotics and Automation Letters Article Video
R. Zufferey, *A. Ortega Ancel, C. Raposo, S. F. Armanini, A. Farinha, R. Siddall, I. Berasaluce, H. Zhu, M. Kovac*

2018 **Power and Control Autonomy for High-Speed Locomotion With an Insect-Scale Legged Robot**.
RA-L, Robotics and Automation Letters Article Video
Goldberg B*, Zufferey R*, *Doshi N, Helbling EF, Whittredge G, Kovac M, Wood RJ*

2017 **A biologically inspired, flapping-wing, hybrid aerial-aquatic microrobot**.
Science Robotics Article Video
Chen Y, Wang H, Helbling EF, Jafferis NT, Zufferey R, *Ong A, Ma K, Gravish N, Chirarattananon P, Kovac M, Wood RJ. A*

Presented Conference Proceedings

2019 **SailMAV: design and implementation of a novel multi-modal flying sailing robot**.
IROS, International Conference on Intelligent Robots and Systems Conf Video
R. Zufferey, *A. Ortega Ancel, [...], and M. Kovac*

2019 **Adaptive Morphology in Aerial-Aquatic Robots**.
AMAM, 9th International Symposium on Adaptive Motion of Animals and Machines Article
R. Zufferey, *S. F. Armanini, A. Farinha, M. Kovac*

2018 **Power and Control Autonomy for High-Speed Locomotion With an Insect-Scale Legged Robot**.
ICRA, International Conference on Robotics and Automation Video
Finalist for Best Paper Overall award
Goldberg B*, Zufferey R*, *Doshi N, Helbling EF, Whittredge G, Kovac M, Wood RJ*

2015 **Cubeth ADCS design, implementation and validation tests**.
66th International Astronautical Congress Article
S. Rossi, A. Ivanov, G. Faure, B. Geissman, J. Amiguet, R. Valceschini, M. Starein, R. Zufferey, *G. Burri*

Posters

2018 **Bio-inspired Aquatic Micro Air Vehicle for Environmental Monitoring and Disaster Relief**.
ICRA, International Conference on Robotics and Automation
Alejandro Ortega Ancel, Raphael Zufferey *and Mirko Kovac*

2016 **An Aquatic Micro Air Vehicle**.
Breaking the Surface
Robert Siddall, Alejandro Ortega Ancel, Raphael Zufferey *and Mirko Kovac*

Communication and Outreach

2020 Invited "Extreme Environment Robotics" talk at UK RAS 4th annual Chapter conference

2019 Invited talk at Sevilla Robotics Vision and Control group

2019 Macau, Session Chair and Marine Soft-robotics workshop

2019 Spain Research Institute (IMDEA) Invited Talk on mobile robots

2017 Science Museum London, Gallery Exhibit on Aquatic Micro Air Vehicle prototype

2017 Science Museum 'Lates' Exhibitor

2017, 2018 Imperial Festival Presenter

2016 Live Robotics demo of the AquaMAV as part the Breaking The Surface conference